



Name: _____ Cohort/Batch: _____ Date: _____ Score: _____

MYSQL PRACTICE SHEET - 10 CONCEPTUAL Q&A

10 conceptual Q&A Databases

TOTAL: 10 QUESTIONS

Q1 What type of database is MySQL and what are its core strengths? [Threat Model](#)

Q6 How do you monitor and tune slow queries in MySQL? [Observability](#)

Q2 How does MySQL guarantee consistency and handle transactions? [Architecture](#)

Q7 What backup and restore strategies does MySQL support? [Recycle](#)

Q3 What index types does MySQL support and when do you use each? [Optimization](#)

Q8 How do you handle schema migrations in MySQL? [Compliance](#)

Q4 How do you design a schema or data model in MySQL? [Hardening](#)

Q9 What security features does MySQL provide? [Testing](#)

Q5 How does MySQL scale horizontally? [SSO / IdP](#)

Q10 What are common anti-patterns when using MySQL in production? [Tuning](#)



ANSWER KEY & EXPLANATORY SOLUTIONS

Comprehensive verified answers, best-practice configs, security controls & reference

VERIFIED SOLUTIONS

A1 MySQL is a relational, document, key-value, graph, Mitigation
MySQL is a relational, document, key-value, graph, or time-series store. Its strengths such as ACID guarantees, horizontal scale, or flexible schema guide the workloads it is best suited for.

A6 MySQL provides a slow query log, explain/explain Observability
MySQL provides a slow query log, explain/explain plan, or profiling tools to surface inefficient queries. Common fixes include adding indexes, rewriting queries, or increasing cache size.

A2 MySQL provides either full ACID transactions or Primitives
MySQL provides either full ACID transactions or eventual consistency depending on its CAP theorem positioning. Transaction support varies from none (simple K/V stores) to serializable isolation (relational DBs).

A7 MySQL supports logical backups (export SQL or Automation
MySQL supports logical backups (export SQL or BSON), physical backups (filesystem snapshot), and continuous replication to a standby. Point-in-time recovery requires WAL/oplog archiving on top of backups.

A3 MySQL offers B-tree, hash, full-text, spatial, or Hardening
MySQL offers B-tree, hash, full-text, spatial, or composite indexes depending on the engine. Choose based on query patterns: B-tree for range queries, hash for equality lookups, full-text for search.

A8 Schema migrations in MySQL are managed with Governance
Schema migrations in MySQL are managed with versioned migration files applied in order by a migration tool. Migrations should be idempotent and tested in staging before production to avoid downtime.

A4 Schema design in MySQL involves normalizing relations Gotchas
Schema design in MySQL involves normalizing relations (relational DB) or denormalizing for access patterns (document/key-value). Understanding query needs upfront prevents costly migrations later.

A9 MySQL supports role-based access control, encrypted Assessment
MySQL supports role-based access control, encrypted connections (TLS), encryption at rest, and audit logging. Always restrict user privileges to the minimum needed and disable anonymous access in production.

A5 MySQL scales via replication (read replicas) for Federation
MySQL scales via replication (read replicas) for read-heavy workloads and sharding (partitioning data by key) for write-heavy ones. Some MySQL implementations handle this automatically; others require manual configuration.

A10 Common mistakes include selecting all columns instead Optimization
Common mistakes include selecting all columns instead of needed fields, missing indexes on frequently queried columns, running migrations without backups, and storing large blobs directly in the database instead of object storage.